

■ End-to-End Stock Forecasting System (Ollama Integration)

1■■ Data Pipeline (Backend ML Workflow)

Module 1: Financial Data Acquisition

- 1 Fetch stock data dynamically using Yahoo Finance API (yfinance).
- 2 Downloads last 5 years of data for the selected MNC (TCS, Infosys, etc.).
- 3 No static CSV files are saved.

Module 2: Data Cleaning & Feature Engineering

- 1 Missing values automatically handled via pandas.
- 2 Optional features for visualization: SMA (10, 50) and Correlation Heatmap.
- 3 Outlier handling and manual preprocessing are not required.

Module 3: Data Preparation

- 1 Normalization performed dynamically using MinMaxScaler.
- 2 Sequence length = 60 days for LSTM predictions.
- 3 No scaler files (.pkl) are saved; all scaling done in runtime.

Module 4: Model Prediction

- 1 ARIMA (5,1,0): Time-series forecasting model on Close prices.
- 2 LSTM: Pre-trained neural network model (lstm_stock_model.keras).
- 3 Forecast horizon = (Selected Future Date – Last Available Date).
- 4 No .pkl models saved; predictions computed directly in memory.

2■■ Streamlit Application (Frontend)

Step 1: User Authentication

- 1 Login required before accessing the app.
- 2 Available users: saurabh / 12345 and admin / admin123.

Step 2: Select Company

- 1 Dropdown menu of 30+ Indian MNCs (e.g., TCS, Infosys, Reliance).
- 2 Fetches 5 years of stock data from Yahoo Finance.

Step 3: Choose AI Model

- 1 Select local Ollama AI model from sidebar: gemma3:4b, mistral, llama2, or None.
- 2 Used to generate short financial insights (not for forecasting).

Step 4: Prediction Setup

- 1 User selects a future date.
- 2 Steps = (future date – last date in dataset).

Step 5: Mode Selection

- 1 Single Model Mode: Choose ARIMA or LSTM for individual predictions.
- 2 Comparison Mode: Runs both ARIMA and LSTM together with side-by-side results.
- 3 Displays tables, % change, trend direction, and charts.

Step 6: Visualization Options

- 1 None → Only predictions.
- 2 Moving Averages → SMA 10 & SMA 50 with closing price.
- 3 Correlation Heatmap → Correlation matrix (Open, High, Low, Close, Volume).

Step 7: AI Chatbot (Ollama Integration)

- 1 Interactive finance chatbot using local Ollama API.
- 2 Understands company names, dates, and user queries automatically.
- 3 Combines historical prices, ARIMA & LSTM predictions, and AI-generated insights.

3■■ Final Outputs

- 1 All predictions, charts, and tables are dynamically generated in Streamlit.
- 2 No intermediate files (.csv or .pkl) are stored.
- 3 Required files in directory: ollama.py, lstm_stock_model.keras.
- 4 Outputs include:
- 5 • Predicted closing price, trend, and % change.
- 6 • Interactive charts for ARIMA, LSTM, and comparison modes.
- 7 • AI-driven financial insights and chatbot responses.